CeylonTourMate.lk - Class Diagram

Admin
(inherited)

+ manageAdminAccounts()
+ viewPendingRequests()
+ approveRejectRegistration()
+ manageUserAccounts()
+ approveRejectListing()
+ viewAllBookings()
+ sendNotification()
+ viewCommissionOverview()
+ changeCommissionRate()

Legend

- attribute + method()

hollow triangle = inheritance (is-a)

open arrow = association, labeled with multiplicity

User
- id: int
- firstname: string
- lastname: string
- username: string
- email: string
- password: string
- phone: string
- profilePic: string
- role: string
- isSuspended: bool
- isApproved: bool
- securityQuestion: string
- securityAnswer: string

+ register()
+ login()
+ logout()
+ editProfile()
+ findAccount()
+ verifySecurityAnswer()
+ resetPassword()

ShopListing
- shopType: string
- openTime: time
- closeTime: time

+ payListingFee()

HotelListing
- starRating: int
- amenities: string
- totalRooms: int
- checkInTime: time
- checkOutTime: time

+ addRoom()

TourGuideListing
- languages: string
- specialization: string
- experienceYears: int
- availability: string

+ setAvailability()

ExperienceListing
- experienceType: string
- duration: string
- maxGroupSize: int
- availability: string

+ setAvailability()

Listing
- id: int
- userId: int
- title: string
- description: string
- location: string
- city: string
- price: decimal
- status: string
- type: string

+ addListing()
+ updateListing()
+ deleteListing()
+ uploadImage()
+ getAverageRating()

Room
- id: int
- hotelListingId: int
- description: string
- roomType: string
- pricePerNight: decimal
- capacity: int
- isAvailable: bool

+ setRoomAvailability()

ListingImage
- id: int
- listingId: int
- imagePath: string

(none)

Booking
- id: int
- touristId: int
- listingId: int
- roomId: int
- status: string
- startDatetime: datetime
- endDatetime: datetime
- guestCount: int
- totalPrice: decimal
- commissionAmount: decimal

+ makeBooking()
+ cancelBooking()
+ confirmBooking()
+ completeBooking()
+ viewBookingStatus()

Payment
- id: int
- userId: int
- shopListingId: int
- bookingId: int
- type: string
- amount: decimal
- paymentStatus: string
- feeMonth: int
- feeYear: int

+ completePayment()
+ payListingFee()

Review
- id: int
- touristId: int
- listingId: int
- bookingId: int
- rating: int
- comment: string

+ writeReview()

Notification
- id: int
- senderId: int
- receiverId: int
- message: string
- isRead: bool

+ sendNotification()
+ receiveNotification()

Commission
- id: int
- rate: decimal
- updatedBy: int

+ changeRate()
+ calculateCommission()

TourGuideLicense
- id: int
- userId: int
- licenseNumber: string
- licenseImage: string

+ submitLicense()

Message
- id: int
- senderId: int
- receiverId: int
- listingId: int
- message: string
- isRead: bool

+ sendMessage()
+ replyToMessage()

Admin → User
ShopListing 1 — contains → 0..* Listing
HotelListing 1 — contains → 0..* Listing
TourGuideListing — Listing
ExperienceListing — Listing
User 1 — creates → 0..* Listing
User 1 — sends → 0..* Notification
User 1 — receives → 0..* Notification
User 1 — updates → 0..* Commission
User 1 — submits → 0..* TourGuideLicense
User 1 — makes → 0..* Booking
User 1 — writes → 0..* Review
User 1 — sends → 0..* Message
User 1 — receives → 0..* Message
ShopListing 1 — pays fee → 0..* Payment
Room 0..* — hotel bookings only → 0..* Booking
Listing 1 — has → 0..* ListingImage
Listing 1 — booked as → 0..* Booking
Listing 1 — receives → 0..* Review
Listing 1 — about → 0..* Message
Booking 1 — paid via → 0..* Payment
Booking 1 — reviewed in → 0..1 Review
Payment 0..* — Review